

DEEP ARCHITECTURAL FLOOR PLAN GENERATION: AN APPROACH FOR OPEN-PLANNED RESIDENTIAL SPACES

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Abstract. This research investigates the collaborative potential of artificial intelligence and deep learning in architectural design, focusing on comprehending and synthesizing the complex relationships within architectural floor plans. The primary question addressed is whether deep learning algorithms can effectively generate residential floor plans characterized by open-planned architectural spaces. To address this, the study introduces a novel model employing generative adversarial networks (GANs) to create open-planned layouts within residential floor plans. Open-planned spaces refers to a design approach in which interior spaces within a structure are intentionally devoid of traditional partitioning elements such as walls and doors. The layout typically features interconnected and visually continuous spaces that flow seamlessly from one area to another. The research contributes by addressing a gap in the literature through the exploration of functional space differentiations within residences characterized by open plan arrangement without walls as a separating element. Furthermore, the study extends this investigation by applying the proposed methodology to angular and circular plans as well as orthogonal plan sets. In the generative model created with GAN, the space functions are defined and

labelled with the RGB color codes assigned to them. For the RGB label representation of the open-plan layout, gradient coloring prepared. By using this method, it was investigated whether the generation of the plans was realized with an open-plan structure by examining the gradient generation results. In the generative model, the footprint of the plan is given as an input for the algorithm to produce by adhering to an outer boundary. Accordingly, it is aimed to learn how the network can be arranged within the given boundaries. The Pix2pix method was used for this generative model, which is defined as the problem of obtaining images from images. The model results advance the AI-driven understanding of architectural design by providing architects with an innovative tool to explore open-plan spatial solutions.

Keywords: floor plan generation, deep learning, generative adversarial networks.

ملخص. تبحث هذه الورقية البحثية في الإمكانيات التعاونية للذكاء الاصطناعي والتعلم العميق في التصميم المعماري، مع التركيز على فهم وتجميع العلاقات المعقدة ضمن مخططات الطابق المعمارية. السؤال الأساسي الذي يتم تناوله هو ما إذا كانت خوارزميات التعلم العميق يمكنها بشكل فعال إنشاء مخططات سكنية تتميز بمساحات معمارية ذات مخطط مفتوح. ولمعالجة ذلك، تقدم الدراسة نموذجًا جديدًا يستخدم شبكات الخصومة التوليدية (GANs) لإنشاء تخطيطات مفتوحة ضمن مخططات الطابق السكنية. تشير المساحات ذات المخطط المفتوح إلى أسلوب التصميم الذي تكون فيه المساحات الداخلية داخل المبنى خالية عمداً من عناصر التقسيم التقليدية مثل الجدران والأبواب. يتميز التخطيط عادةً بمساحات مترابطة ومستمرة بصرياً تتدفق بسلاسة من منطقة إلى أخرى. يساهم البحث من خلال معالجة فجوة في الأدبيات من خلال استكشاف الفروق المكانية الوظيفية داخل المساكن التي تتميز بالترتيب المفتوح بدون جدران كعنصر فاصل. علاوة على ذلك، توسع الدراسة هذا البحث من خلال تطبيق المنهجية المقترحة على المخططات الزاوية والدائرية بالإضافة إلى مجموعات المخططات المتعامدة. في النموذج التوليدي الذي تم إنشاؤه باستخدام GAN، يتم تعريف وظائف المساحة وتسميتها برموز ألوان RGB المخصصة لها. لتمثيل تسمية RGB للتخطيط المفتوح، تم إعداد تلوين متدرج. باستخدام هذه الطريقة، تم التحقق مما إذا كان إنشاء الخط قد تم تحقيقه بهيكل مفتوح من خلال فحص نتائج توليد التدرج. في النموذج التوليدي، يتم تقديم بصمة الخطة كمدخل للخوارزمية لإنتاجها من خلال الالتزام بالحدود الخارجية. وبناءً على ذلك، يهدف إلى معرفة كيفية ترتيب الشبكة ضمن الحدود المعطاة. تم استخدام طريقة Pix2pix لهذا النموذج التوليدي الذي يعرف بأنه مشكلة الحصول على الصور من الصور. تعمل نتائج النموذج على تعزيز الفهم القائم على الذكاء الاصطناعي للتصميم المعماري من خلال تزويد المهندسين المعماريين بأداة مبتكرة لاستكشاف الحلول المكانية ذات المخطط المفتوح.

الكلمات المفتاحية: توليد المساقط الأفقية، التعلم العميق، شبكات الخصومة التوليدية.

1. Introduction

A floor plan can be defined as a bi-dimensional representation of the spatial configurations and dimensions of spaces. Different spatial configurations define different spatial boundaries, leading to a unique spatial experience. The complex relationships represented in architectural plans are mainly based on dynamic interactions of perceiving and traversing spatial spaces. The accessibility provided by spatial functions are supported by topological relationships that govern the flow and connection of architectural spaces. In addition, the visual interaction between spaces is realized through openings such as windows and galleries, creating an interaction of vision and visibility. It is noteworthy, however, that factors such as visual relationships that determine the experiential quality of architectural spaces are often not included in architectural plan-generating processes.

Models used in architectural plan generation are mostly based on defined topological relations, constraints such as not overlapping spaces, leaving undefined areas, and/or objectives such as minimizing the circulation area. Topological relations represent architectural units that should be close to each other depending on the building organization. This relationship is based only on the physical proximity-distance relationship between spaces. Lobos and Donath (2010) divide architectural plan generation criteria into measurable rational criteria and unmeasurable design criteria. While rational criteria can be counted as sun, landscape, accessibility, proximity of related functions, minimizing circulation areas, unmeasurable design criteria are defined as geometric composition, form formed in third dimension, aesthetics of form / golden ratio etc. Existing models are inadequate of specifying whether the spaces are separated by a direct opening, a door, a solid wall surface, or a surface that is transparent like glass and allows to see/be seen. While a transparent wall surface does not prevent the act of seeing, it limits the act of going. A solid wall surface creates both a visual and a physical boundary. For this reason, the definition of topological spatial relationship based only on the proximity-distance relationship is insufficient to represent the architectural space. In architecture, spatial divisions can be defined as separate rooms separated by physical walls. However, one of the basic principles of architecture is that the use of space within a building should be flexible or flexible (Jabi et al., 2018). For this reason, the topological relations of the open plan layout that allows multiple uses is important.

The aim of the research is to investigate whether deep learning algorithms can make inferences about the visual relationship of spaces from the training sets, and the visual and physical relationship required by the architectural space function (while spaces such as bedrooms and bathrooms have less visual and

physical relationships due to the privacy situation, common use spaces such as living rooms have higher) or not.

As Stiny (2006) mentioned “Spatial relations for a given vocabulary sometimes allow its shapes to interpenetrate in designs. Interpenetration like spatial ambiguity can be a valuable technique in design.” Spatial ambiguity is a quality of space that an architect can deliberately incorporate into the building design while providing a subjective experience from the user's point of view. The concept of flexibility, a product of the Modern Movement in Architecture, gained importance after the 1950s, adding the elements of "time" and "unknown" to the design and brought a new breath to functional architecture. According to Collins (1968), “flexibility is a kind of functionalism” and he defines flexibility as a closed circuit that is specified by the architect, specialized for not one but more than one configuration. Spatial ambiguity occurs in open, flexible plan designs where the spatial configuration does not reveal clear and precise functional distinctions. In this study, it is suggested to create floor plan images with gradient coloring for the flexible arrangements in the plan and the visual relationships resulting from it.

The image consists of the combination of more than one pixel. The whole of the techniques whose input is image and output is information in different modalities is called "computer vision". The output of computer vision techniques is semantic information related to the input image or derived from the input images. For the recognition and the generation of the floor plan visuals, it is important to learn the semantic information contained. Floor plan recognition is different from other separation methods. Because the elements that make up the plan are not only found singularly, but the relationship also they express with each other and the spaces they represent are important. Computerization and, in particular, computer vision presents a new opportunity for expressing the general cognitive mechanisms and the domain knowledge contained in these representations. It is a powerful method used on visual data and has good potential for recognizing and generating architectural drawings. Therefore, architectural housing plan design as a complex and ill-defined problem is investigated with deep learning methods, which is a sub-branch of artificial intelligence. The paper presents a novel model that aims to generate open-planned architectural spaces represented in residential floor plans with generative adversarial networks (GANs) which is one of the deep generative networks.

2. Related Works

The methods used in the architectural plan generation can be listed as shape grammars, genetic algorithms (GA), general probabilistic methods, growth

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models and deep generative models. In deep learning, neural networks are not structured according to rules or previous algorithms, they learn patterns/patterns with training sets. This contrasts with previous generative design approaches. For example, in shape grammars, the initial form and the rules to be applied repeatedly are determined by the designer. In deep learning, on the other hand, the rules for combining shapes and forms are not predetermined, the algorithm discovers them from the training data.

In the plan generation and optimization studies made with genetic algorithm, the process of finding the most suitable ones for the fitness function or more than one conflicting function from the solution set in a research space is followed. As an early example of architectural space arrangement, Rodrigues, Gaspar, & Gomes (2013) created an optimization model by considering contiguity, space overlap, span overlap and orientation, floor dimensions and overhang. They applied this framework to the housing plan generation problem to validate the proposed method. In another example of a residential setting, they used GA to maximize symmetry, structural safety, stair connectivity, and the amount of daylight exposure of the facades (Song et al., 2016). In GAs, the fitness function is calculated over the defined design variables. Therefore, unlike deep learning, these functions and variables are determined by the designer.

General probabilistic approaches have explored the application of Bayesian networks (Merrell et al., 2010) and Markov chains (Swahn, 2018) to learn from sample images in the synthesis of the architectural plan.

In cellular automata, which is one of the growth models, cells arranged on regular Cartesian grids in two or three dimensions are loaded with predefined states with neighborhood relations that differ depending on the rules (Dincer et al., 2014).

Deep neural networks mimic the inner workings of the human brain and consist of (1) an input layer, (2) hidden layers (which process data), and (3) an output layer that produces results. It can be reconfigured, and the weights of neurons can be manipulated iteratively to improve its performance. In this training process, DNN (Deep Neural Networks) learns to get the right output in the process.

As et al. (2018) proposed a generation model with deep artificial neural networks, using sets of architectural plans represented as “graphs” along with the livability and adaptability criteria. The algorithm proposes a new architectural plan with orthogonal designs by combining the subgraphs. However, the limited sample dataset and subjective evaluation processes restrict its general applicability.

Alternative approaches have been explored, including recognizing architectural plans from color-labeled images (Huang and Zheng, 2018). The algorithm demonstrates a strong capability in identifying spaces with well-

defined boundaries and distinct features like bedrooms, kitchens, and toilets. However, it encounters challenges when distinguishing spaces where boundaries converge, as seen living rooms and corridors. The algorithm does not propose a new plan with different topological relations but aims to reach the architectural plan representation output from the image represented by color labels. Notably, it did not perform well with non-orthogonal plans due to the dataset's dominance of orthogonal plans.

Wu et al. (2019) introduced DeepLayout, focusing on residential architectural plans. It relies on predefined room connections and layouts, with user interaction options, but lacks circulation area definition. DeepLayout had challenges in placing sufficient space and even privacy issues such as the transition from the bathroom to the balcony emerged. Additionally, Hu et al. presented Graph2Plan, a learning-based approach that adapts similar layouts to given input boundaries. Both DeepLayout and Graph2Plan require predefined rules for door and window placement, whereas our approach learns these rules from data.

Nauata et al. (2020) propose HouseGAN, a graph-constrained generative adversarial network, to generate floor plans based on room types and spatial relationships. It encountered issues with room sizes, shapes, placement, and inaccessible spaces. The model does not consider space sizes, only assigning rectangular forms to spaces without addressing doors and windows. FloorGAN (Abhinav et al., 2023) designed to create floor plans while adhering to user-defined constraints via graphs. This approach incorporates user input, including room types and spatial relationships, to generate layouts that comply with these specifications.

All these mentioned models deal with the issue of generating floor plan either as a topological relationship, and (or) simply a coloring problem according to a function definition only. However, a floor plan is not just a visual combination of architectural elements such as walls, doors, windows, furniture. It is a visual representation of how they relate to each other and how they are arranged to create different types of spaces and experiences. Therefore, the presented research does not limit itself to rectangular plans only; expands its floor plan generation reach to angled and curved designs. The importance of visual relationships as well as topological relationships in the process of creating architectural plans is also emphasized by working on open-planned architectural plans. Furthermore, the primary objective of this research is to unleash the algorithm's creative capabilities, allowing it to generate original designs without imposing strict constraints dictated by a mere graph.

3. Methodology

In the proposed generative model, it is aimed to make the application in angular and circular plans as well as orthogonal plan sets, and to consider the functional area differences in the residential plans with an open/flexible plan arrangement without walls within the scope of the model. In line with these goals, gradient coloring is suggested for RGB label representation of flexible layout. Recoloring studies on R3D dataset were carried out according to the proposed gradient coloring method. In this way, the gradient generation outputs are explored as a result of the generative model. Generative adversarial networks are selected for this model. GANs, learn statistically significant phenomena from the data presented to them without the need for labels and produce visually realistic images. GANs are generative models that learn to map an output image from a random noise vector (Goodfellow et al., 2014). In this section, the proposed gradient coloring method and the network architecture used for the preparation of the data set are explained in this section.

3.1. PREPARATION OF DATASET AND LABELING PRINCIPLES

The data set used in this study is R3D data set which is prepared by Zeng et al. (Zeng et al., 2019) with R2V data set. R2V mostly consists of rectangular shaped plans with uniform wall thickness images from Raster-to-Vector (Liu et al., 2017). A total of 182 plan images were used, 131 of which had an open plan layout. For R3D, rounded-shaped floor plans were added to the dataset by Zeng alongside the original images in Rent-3D (Liu et al., 2015). 112 of the 194 plan images have an open plan layout. Compared to R2V, R3D has non-orthogonal plan sets with nonuniform wall thickness (Zeng et al., 2019) (Fig.1). While 168 of the images selected from the R2V plan set are orthogonal plans, this number is only 32 in R3D. Both data sets are open-source sets that can be accessed from given references.



Figure 1. Comparison between the R2V (left) and R3D (right) data sets (Zeng et al., 2019).

In these datasets, the same label is used for some open-planned zones, such as the living room and the dining room, that generally do not have separating walls and are mostly adjacent to each other. In this study, we used different labels to separate the connected room zones, even though they are located next to each other without walls separating them (Fig. 2). Due to the spatial ambiguity of open-plan spaces, it is not clear exactly where space functions begin and end. Gradient coloring representation is suggested for the representation of such spaces with RGB color labels. A gradient is a path from one color to another in a color space. Just as a line segment can go from (x_1, y_1, z_1) to (x_2, y_2, z_2) in 3-dimensional space, so a color be reached from $rgb(r_1, g_1, b_1)$ to $rgb(r_2, g_2, b_2)$ in RGB space. The gradient transition does not always have to be linear. Gradient color transitions can be created in a free form determined by points as well as linear, radial, or reflected. Each sub-gradient color ($c(n)$) is calculated for each pixel transition on the gradient path.

Where the starting color (c_1) of the gradient is defined as $rgb(r_1, g_1, b_1)$, the end color (c_2) of the gradient is defined as $rgb(r_2, g_2, b_2)$ and p is the total number of pixels of the image.

When the gradient process on an image proceeds incrementally in equal intervals, the first step in computing the sub-gradient color is taking the difference between the R, G, and B channels of the colors. Then this difference is divided by the total number of pixels for which the gradient will be created. Each of these values are added to the R, G, and B channels for each subsequent transition in the gradient as shown in Eq.1.

$$c(n) = rgb(r_1 + \left(\frac{r_2-r_1}{p}\right) * n, g_1 + \left(\frac{g_2-g_1}{p}\right) * n, b_1 + \left(\frac{b_2-b_1}{p}\right) * n) \quad (1)$$

However, in a space with an open plan arrangement, the spatial distribution of functions is not always equal. For example, in the distribution of the living room and the dining room in a space, the living area usually occupies a percentage of more space.

For this, the distribution scale of the spatial function change is determined and the pixel values in the distribution range where this change occurs are in the middle transition color (c_{avg}) of these two colors. It is defined as $rgb(r_{avg}, g_{avg}, b_{avg})$ and calculated by taking the arithmetic average of each R, G, and B channels of the starting and end color (Eq.2).

$$\begin{aligned} c_1 &= rgb(r_1, g_1, b_1) \\ c_2 &= rgb(r_2, g_2, b_2) \\ c_{avg} &= rgb(r_{avg}, g_{avg}, b_{avg}) = rgb\left(\frac{r_1+r_2}{2}, \frac{g_1+g_2}{2}, \frac{b_1+b_2}{2}\right) \end{aligned} \quad (2)$$

The pixel location of each subsequent transition is important for the definition of the transition state between spaces. Calculation of subsequent colors interval positions differ depending on whether the range field (x) is

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before (closeness to the starting color) or after (closeness to the end color). Number of pixels on either side of the range field is calculated by multiplying the inter-functional distribution scale value of an open space (s) with the total number of the pixels. Accordingly, the color change between pixels is calculated as denoted in Eq.3.

if $s.p \leq x$ then

$$c(n) = rgb\left(r_1 + \frac{r_{avg}-r_1}{s.p}, g_1 + \frac{g_{avg}-g_1}{s.p}, b_1 + \frac{b_{avg}-b_1}{s.p}\right) \quad (3)$$

$$\text{else } c(n) = rgb\left(r_{avg} + \frac{r_2-r_{avg}}{(1-s).p}, g_{avg} + \frac{g_2-g_{avg}}{(1-s).p}, b_{avg} + \frac{b_2-b_{avg}}{(1-s).p}\right)$$

The suggested gradient coloring method was applied to these plans by using Adobe Photoshop and Illustrator to manually recolor spatial differences in open-plan areas. Table 1 shows the detailed colors and RGB codes used.

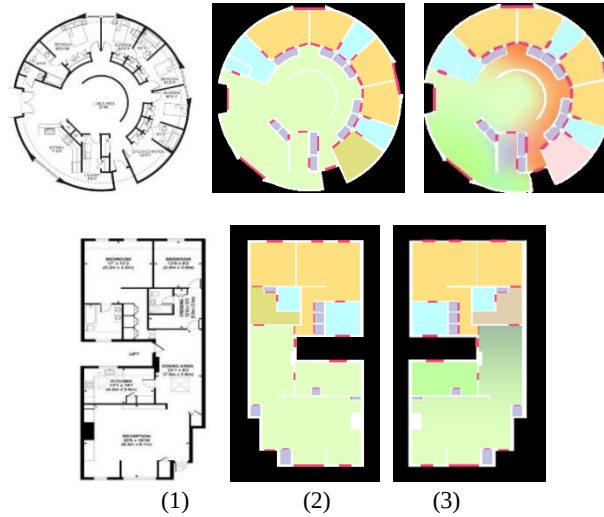


Figure 2. (1) Floor plan (2) R3D data set coloring (3) Proposed gradient recoloring in open-planned spaces ID.3 (top) – ID. 45737995 (bottom)

TABLE 1. RGB code and representations of each usage label.

Space usages	RGB Code and RGB color representation
Living room (LivR)	224, 255, 192
Kitchen (Ktch)	192, 255, 160
Dining room (DinR)	160, 192, 160
Bath / WC (BthR)	192, 255, 255
Bedroom (BedR)	255, 224, 128
Dressing room	224, 208, 176
Study room (StdR)	255, 224, 224
Storage	192, 192, 224
Corridor / hallways (Hall)	255, 160, 96
Courtyard	192, 192, 192
Fireplace	224, 224, 224
Windows/doors	255, 60, 128
Wall	255, 255, 255

In addition, even if two separate spaces are separated by a wall, if they are defined by the same color, these spaces are also separated. For example, with the ID 45665893 plan, both the living room and the kitchen are represented in light green. The kitchen space has been recolored with dark green (Fig. 3).

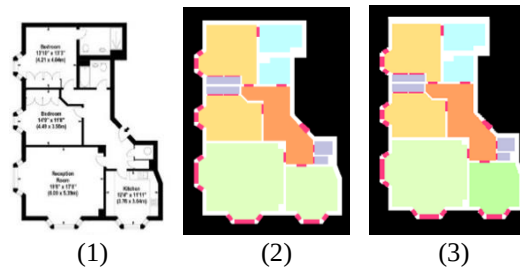


Figure 3. (1) Floor plan (2) R3D data set coloring (3) Proposed recoloring in different spaces separated by a wall ID.45665893

The walls are defined with white color (255, 255, 255) and the remaining area with black (0, 0, 0) so that the computer does not spend too much calculation time in the surrounding visual areas where there is no plan image. In the generative model, it is thought that the footprint of the plan in the field is given as input in order that the algorithm produces by adhering to an outer boundary. Accordingly, it is aimed that the deep network learns how to arrange a plan within the given boundaries (Fig. 4). The prepared data set was prepared

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with input and output images of 512x256 pixels. Then, the image is divided into two, half of which is 256x256 pixels as input and the other half of 256x256 pixels is processed as output label.

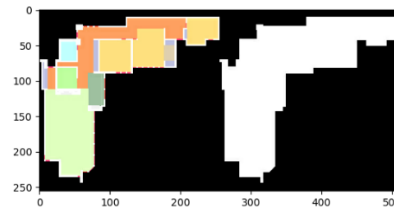


Figure 4. Input and output image prepared for the generative model ID.5

3.2. NETWORK STRUCTURE

The Pix2pix (Isola et al., 2017) network was used for the generative model, which is defined as the problem of obtaining images from images. Pix2pix is one of the conditional adversarial networks used in image-to-image conversion. These networks learn to map from input image to output image. This approach has been shown to be effective in synthesizing images from tag maps, reconstructing objects from edge maps, and coloring images. It consists of two main artificial neural networks: Generator and Discriminator.

The discriminative network is trained to recognize the images in the training set. In the generative network, a footprint image of a floor plan was used as input and it is aimed to generate new visuals as closely as possible to the visuals in the training set. The discriminator tries to distinguish whether the images from the generator are real or fake (Fig.5). As the generative network evolves, it learns to generate more lifelike images, while the discriminating network can differentiate between the actual image and the generated one. The development of the generative network occurs as a result of feedback according to the error values obtained with the loss function of the discriminator (Goodfellow et al., 2014). For this type of competitive training, the goal is to achieve a Nash equilibrium (Heusel et al., 2017) where the generated images are indistinguishable from real images.

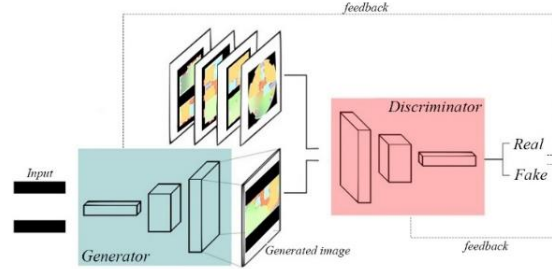


Figure 5. Diagram illustrating the process of Pix2Pix network.

Combination of some deep learning techniques is described to train GANs. These techniques include: (i) fully convolutional network and (ii) Batch Normalization (BN). In the convolutional network, filtering operations are used to increase and decrease the spatial dimensions of the image. BN normalizes feature vectors to have zero mean and unit variance across all layers. This helps stabilize learning and cope with poor weight initiation problems (Radford et al., 2015). The steps performed in this study are represented in the network architecture diagram with the number of channels (Fig. 6). The down-sampling and up-sampling in generator networks consist of 4×4 convolution filters with stride of 2. It is followed by a BN and rectified linear unit (ReLU). In activation function of ReLU, if a value is negative, it passes 0, if it is positive it passes as it is (Zhao et al., 2020). It behaves like human neurons; they do not respond to signals below a certain level. In spite of this, the discriminator down-sampling step uses the Leaky ReLU function. The difference of Leaky ReLU from ReLU is that instead of resetting negative values, it returns a very small value. With this situation, it is aimed to prevent dead neurons.

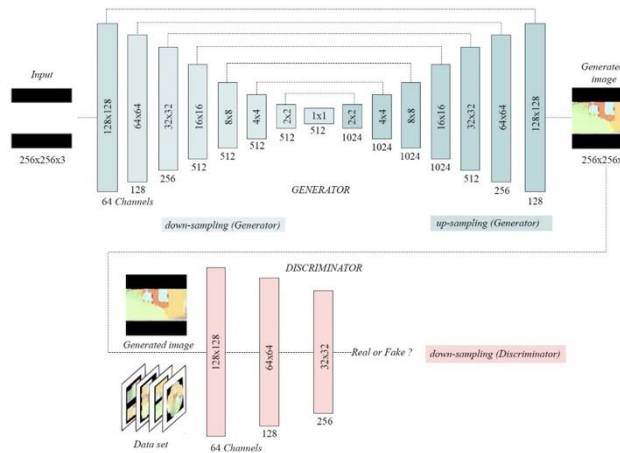


Figure 6. Diagram illustrating the network architecture of Pix2Pix.

4. Results

Three distinct input image categories were selected to facilitate a comprehensive comparative analysis of generative outcomes obtaining to the model trained using the R3D dataset. The primary input form includes an assessment of generative outputs within the context of orthogonal scenario, as shown in Figure 7. On the other hand, the input of the second generative model were chosen to generate angular configurations as in Figure 9. Lastly, a circular input image was selected to evaluate the efficacy of floor plan generation in accommodating curvilinear forms, as presented in Figure 11. All training and generating operations are performed using NVIDIA GeForce RTX 2070 GPU. The model prints the image on the screen after 10 epochs and stops the progress of the model after 190 epochs. It takes between 200-300 seconds to complete each epoch steps. It takes approximately 10 hours for the model to complete the process.

From the 30th epoch, the recessed orthogonal plan, started to be formed around the corridor placed in the lower juncture corner. The upper horizontal rectangular section has been designated as a common living area consisting of a living room and an open kitchen since the early epochs. The bathroom is adjacent to the common living area and the bedroom, connected to the corridor (Fig. 7).

L1 loss (defined as the least absolute deviations) is calculated between the images generated by the model and the actual images. This loss function should be as small as possible to allow the generator network to fool the discriminator. At the end of the training, the generator loss was 0.74, and the discriminator loss was 0.62. Increasing this distance means that the generated images move away from the images in the training set, that is, the generating network is not trained.

At the center point of the angular input image, the gradual emergence of a corridor occurred. Subsequently, at the 60th epoch, the establishment of the linkage between the corridor and the exterior wall was created. In general, a large bedroom was positioned along the angled facade of the form. Notably, there were attempts by the algorithm to divide this room in the process; however, complete wall formation was not developed (*epoch 150*). Along the same façade, in the final epoch, a master bathroom connected to the bedroom was generated next to the large bathroom. Conversely, on the orthogonal side, a spacious living area and an open kitchen were designed (Fig.9). Although some undefined wall and discontinuous space formations are observed, it can be interpreted that a generally defined floor plan has been generated.

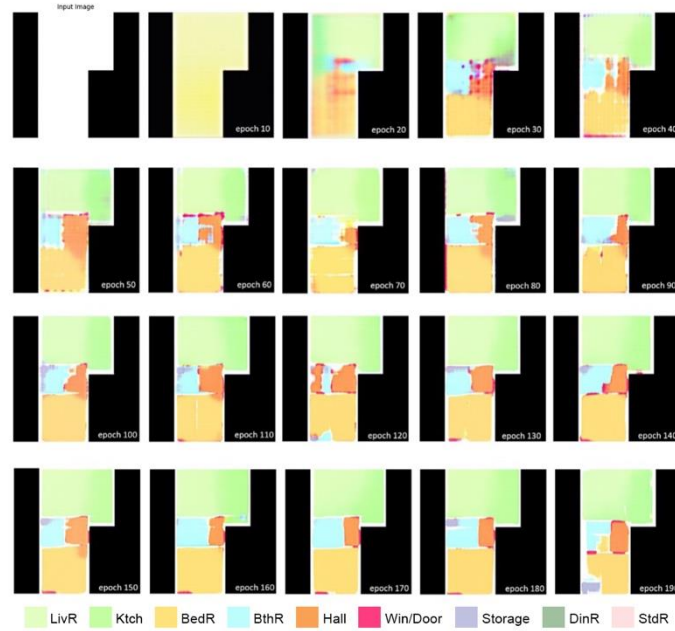


Figure 7. Generative results of orthogonal form with colored label descriptions representing space function

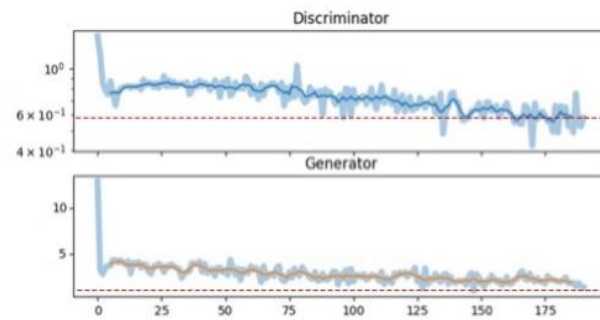


Figure 8. Loss function values of discriminator and generator networks of orthogonal model

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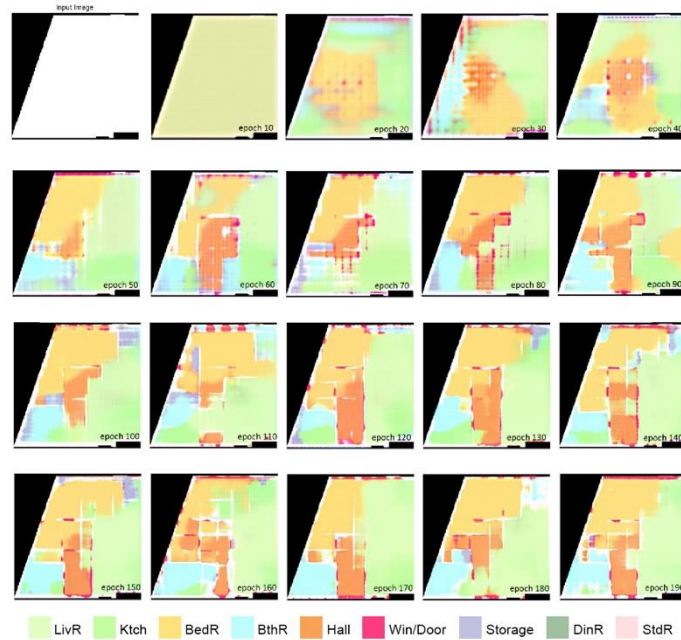


Figure 9. Generative results of angular form with colored label descriptions representing space functions

As determined by the computed loss functions, noticeable reductions in loss values were observed within both the generator and discriminator networks, indicative of progressive learning throughout the process. The discriminator loss was 0.55, while the generator loss was recorded at 1.66. Plots of the loss functions of the generator and discriminator networks during the procedure are visually presented in Figure 10.

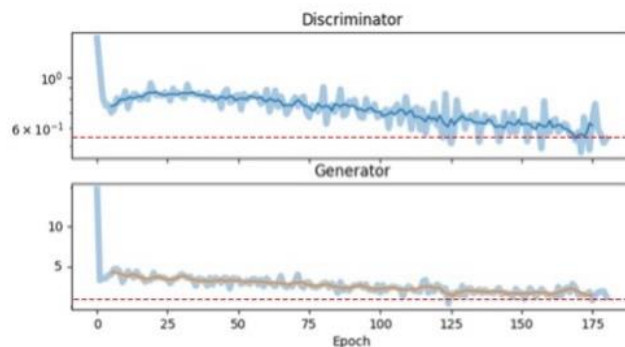


Figure 10. Loss function values of discriminator and generator networks of angular model

In the model, where an input image with a curvilinear form is selected, a corridor space positioned near the center of the form is observed during the production processes. Sleeping units, living room, kitchen and bathroom units are located around this space. Unlike other models, it is observed that more than one bedroom unit is generated. Another notable difference is the generation of the dining room space between living room and kitchen. In addition, color generations representing the study room unit were carried out in the last epochs. There are some problems such as loss of corridor continuity with the intervening storage unit (Fig 11). The discriminator loss achieved a measure of 0.71, while the generator loss was recorded at 1.89 (Fig.12)

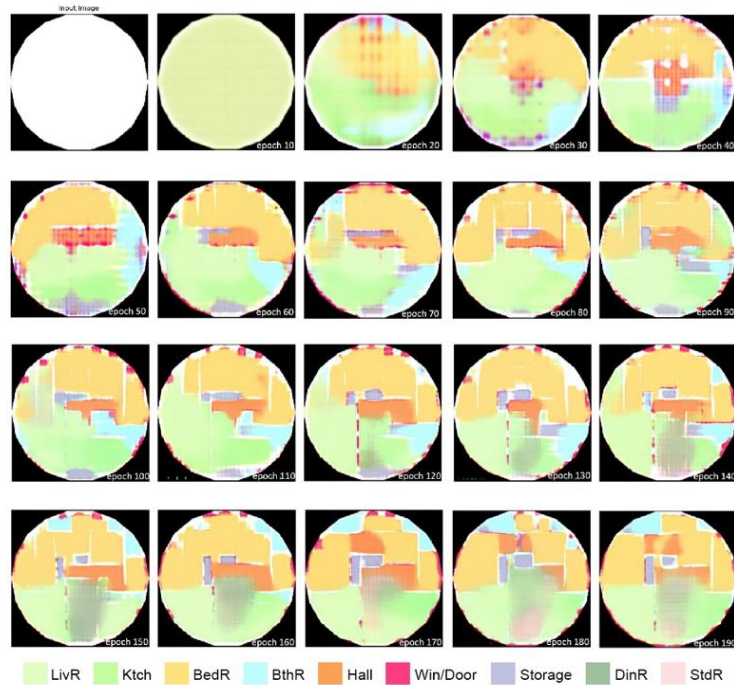


Figure 11. Generative results of circular form with colored label descriptions representing space functions

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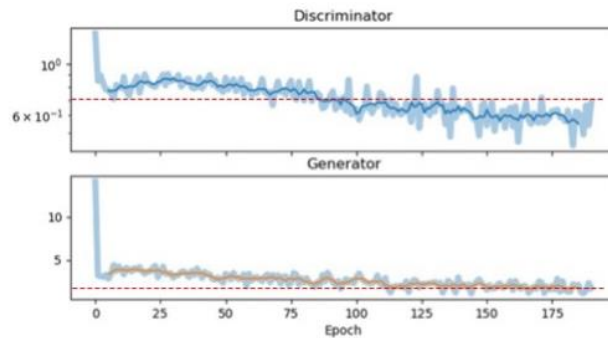


Figure 12. Loss function values of discriminator and generator networks of circular model

5. Evaluation and Discussion

The less deterministic and more holistic character of artificial intelligence methods undoubtedly provides an advantage for its use in architecture. This paper presents case studies for multiple forms on the pixel-based architectural floor plan generation model. Although there are many accepted GAN algorithm evaluation methods in the literature, it is still a question to be answered whether those methods can be used in the evaluation of the outputs consisting of architectural floor plan schemes. Regarding the evaluation methods, it is concluded that if the performance of the GAN algorithm is efficient, the results are also efficient; however, evaluating the generations is a different statement from evaluating the performance of the algorithm for architectural design problems. It should be discussed whether the efficient performance of the algorithm in the generation of architectural plans means that it always produces an acceptable architectural floor plan or not. This is because floor plans contain semantic information that represents many spatial relationships.

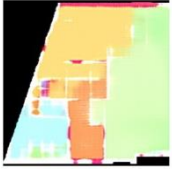
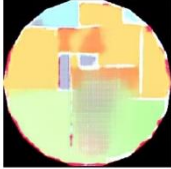
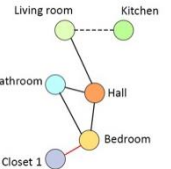
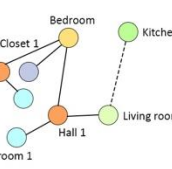
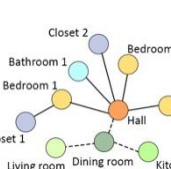
When the result images are examined, it is seen that all the places that should be included in the plan, such as the living room, kitchen, bedroom, bathroom, are present in the generated images. The outcomes were analyzed in terms of function-area distribution and were graphed in percentage terms (Fig. 13). Consistently, the living room and bedroom spaces occupy the most substantial spatial proportions, aligning with expected conventions. Minimizing the circulation area is one of the important objectives in the floor plan generation problem and it is noteworthy that unnecessarily long corridors were not observed in the generation outcomes. Particularly, the angular plan exhibited the highest circulation area among the various configurations considered. This highlights the importance of considering architectural geometry and layout in the analysis of circulation areas in different floor plan configurations.



Figure 13. Function-area distribution analysis of generated floor plans

Evaluating the topological relationships of the floor plan areas generated by the algorithm is a multifaceted task that includes evaluating the connectivity, adjacency and spatial relationships between different functional areas within the layout. The output images have been transformed into a graphical representation where rooms and spaces are nodes and the connections between them (doors, openings, corridors) are edges (Table 2). The plans were evaluated comparatively according to different criteria determined. These can be listed as entrance recognition, natural lighting, spatial distribution logic, circulation efficiency and proportion of spaces.

TABLE 2. Evaluation of the topological relationships of the generated floor plans with comparative analysis criteria

			
	Orthogonal	Angular	Circular
Criteria			

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Entrance recognition	There is no door representation in hall	There is entrance door representation connected to hall	There are doors/windows that can define the open access from living room
Natural light and ventilation	There is no window/door representation in outer edge of the plan	There are large windows only in the bedroom. Missing in common area and bathroom	There are large windows in common areas such as the living room and kitchen. Window-colored pixels are also observed on the exterior of bedrooms
Spatial distribution logic	Common spaces and private spaces are separated from each other	The connection between the kitchen and the bedroom is problematic, there must be a wall in between.	The upper half of the geometry can be interpreted as private, and the lower half divided by common areas
Circulation efficiency	Circulation was minimized with the corridor placed in the center of the geometry.	Access to all spaces is provided through the corridor connected from one edge center of the geometry.	A winding corridor was formed due to the geometric form and 3-bedroom configuration, continuity interrupted by closet
Proportion of spaces	The spaces seem to be of appropriate size.	The bedroom looks too large. It can be partitioned.	The bathroom represented at the top right may be too small for its use function.

6. Conclusion

Spatial configurations represent the dynamic interaction of spaces where they define unique spatial experiences. In the architectural context, spatial ambiguity is considered an architecturally intentional quality that enhances subjective user experience, especially in open, flexible plan designs. While existing models are primarily based on defined topological relationships, although topological relationships based on physical proximity are very important, especially in open, flexible plans that allow multiple uses, they alone do not fully reflect the complexity of the architectural space. In the study, it was shown that deep learning algorithms used visual relations that are vital for architectural functions and whether it can decipher visual field relationships from training data, including physical relationships.

The novel approach involves creating floor plan images with gradient coloring, capturing the essence of flexible spatial arrangements and the resulting visual relationships. Leveraging computer vision, a subfield of artificial intelligence, the research proposes a new generation model by using the opportunity to express the cognitive mechanisms and domain knowledge

found in architectural representations. It is undeniable that pixel-based generative models have limitations regarding the precision in image output. The limitations arise due to the inherent discrete nature of pixels. The output generated by pixel-based models is reliant on the granularity of the pixel grid, and this limitation impacts the capacity to accurately render fine details and nuanced features present in architectural drawings. However, the capabilities of generative deep learning algorithms allow architects to quickly explore numerous design variations and configurations that may not be practical to consider manually. These models should not be seen as "digital designers" but rather as an artificial assistant to support the designer. As a conclusion, while pixel-based generative models are valuable tools for exploring broad design possibilities, their limitations in faithfully generating precise architectural details and complex spatial relationships should be acknowledged.

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